

Package ‘StablePopulation’

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Title Reconstructs Stable and Stationary Survivorship Profiles

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Description Reconstructs age-specific survivorship profiles under a stable and stationary population assumption using a Weibull survival model. Given an age-specific fertility schedule, the package generates Weibull profiles that satisfy the demographic constraint $R_0 = \sum(lx * mx) = 1$, scans candidate beta values, optionally selects the constrained profile that best reproduces an observed survivorship schedule, and derives stable age structures and mortality profiles. It also provides a free two-parameter Weibull fit as a descriptive reference and separate Excel workflows for the historical and extended routes. Methods are based on Martín-González et al. (2019) [doi:10.1016/j.jasrep.2019.03.031](https://doi.org/10.1016/j.jasrep.2019.03.031).

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alpha_objective	<i>Objective Function for uniroot: Finds the Difference Between Births and 1</i>
-----------------	--

Description

This function calculates the difference between the number of births, as calculated with the given values of alpha, beta, and fertility_rates, and the target value of 1.

Usage

```
alpha_objective(alpha, beta, fertility_rates)
```

Arguments

- alpha A numeric value representing the alpha parameter.
- beta A numeric value representing the beta parameter.
- fertility_rates A numeric vector containing the fertility rates.

Details

Typically used as the objective function in root-finding algorithms such as [uniroot](#), to determine the value of alpha that results in exactly one birth.

This function depends on [calculate_population](#), which must be available in your package namespace.

Value

A numeric value giving the difference between the number of births (as calculated) and 1.

See Also

[uniroot](#)

Examples

```
# Basic usage
alpha_objective(0.5, 1.2, c(0.2, 0.3, 0.5, 0.4))

# Example with uniroot:
fertility_rates <- c(0.2, 0.3, 0.5, 0.4)
beta <- 1.2
res <- uniroot(
  alpha_objective,
  interval = c(0.000001, 100),
  beta = beta,
  fertility_rates = fertility_rates
)
res$root
```

calculate_population	<i>Calculates the population for each age group</i>
----------------------	---

Description

This function calculates the population for each age group and the number of births.

Usage

```
calculate_population(alpha, beta, fertility_rates)
```

Arguments

alpha	A numeric value representing the scale parameter (α) of the Weibull distribution. Note: In this context, alpha controls the horizontal scaling of the survival curve.
beta	A numeric value representing the shape parameter (β) of the Weibull distribution. Note: Beta controls the shape of the survival curve (e.g., aging or failure rate).
fertility_rates	A vector of fertility rates for each age group.

Value

A list with the following elements:

population A numeric vector giving the population size for each age group.

births A numeric value giving the total number of births.

Examples

```
calculate_population(0.5, 1.2, c(0.2, 0.3, 0.5, 0.4))
```

derive_demographic_profile

Derive stable-structure and mortality profiles from survivorship

Description

Converts a survivorship profile l_x into the quantities used by the demographic-to-ecological bridge in Model-V2022: the normalized stable structure R_x , the exit-by-death profile D_x , its relative version, and conditional survival between consecutive classes.

Usage

```
derive_demographic_profile(lx, fertility_rates = NULL, tolerance = 1e-08)
```

Arguments

<code>lx</code>	Numeric survivorship profile. It must start at one and be non-increasing.
<code>fertility_rates</code>	Optional fertility schedule with the same length as <code>lx</code> . When supplied, the result also includes $l_x m_x$ and R_0 .
<code>tolerance</code>	Positive tolerance used to validate <code>lx</code> and assess $R_0 = 1$ when fertility is provided.

Details

The stable structure is $R_x = l_x / \sum_x l_x$. The raw exit-by-death profile is $D_x = R_x - R_{x+1}$ for all but the last class, and $D_n = R_n$. Its total equals R in the first class, so it is not generally a probability vector. The relative profile is $D_x / \sum_x D_x$; it is a distinct quantity that sums to one.

Conditional survival is $B_x = l_{x+1} / l_x$. It is returned as NA when the denominator is zero, because the conditional ratio is then undefined. If fertility is supplied, the result also contains $l_x m_x$, the resulting R_0 , and a numerical stability check.

Value

A list of class "stable_population_demographic_profile". Main elements are `stable_structure` (also `R`), `mortality_profile` (also `D`), `mortality_profile_relative` (also `D_relative`), `conditional_survival` (also `B`), `lxmx`, `R0`, `checks`, and a row-by-age table.

See Also

[reconstruct_population\(\)](#), [select_beta\(\)](#), and [scan_beta\(\)](#) to obtain survivorship profiles before deriving these quantities.

Examples

```
mx <- c(0, 0, 0.30, 0.75, 0.60, 0.20)
reconstruction <- reconstruct_population(mx, beta = 1.10)

demographic_profile <- derive_demographic_profile(
  lx = reconstruction$lx,
  fertility_rates = mx
```

```
)
demographic_profile$table
```

find_alphas	<i>Function to find the value of alpha</i>
-------------	--

Description

This function finds the value of alpha using the uniroot method for a given beta and a vector of fertility rates. If the function values at the interval ends do not have opposite signs, it returns the closest value to 0.

Usage

```
find_alphas(beta, fertility_rates, tol = 1e-22)
```

Arguments

beta	A numeric value representing the beta parameter of Weibull distribution.
fertility_rates	A numeric vector containing the fertility rates.
tol	A numeric value representing the tolerance for the uniroot method. Default is 1e-22.

Value

A numeric value giving the estimated value of alpha, either found by uniroot or selected as the endpoint closest to zero if the root is not bracketed.

Examples

```
find_alphas(1.2, c(0.2, 0.3, 0.5, 0.4))
```

find_stablepopulations_root	<i>Find Root Directory of StablePopulation</i>
-----------------------------	--

Description

This internal function searches for the root directory of the StablePopulation project by looking for a folder named StablePopulation in the current or parent directories. It is used internally to locate project-specific files.

Usage

```
find_stablepopulations_root()
```

Value

A character string with the full path to the StablePopulation directory if found. If not found, an error is raised.

fit_weibull_free	<i>Fit a free two-parameter Weibull survivorship model</i>
------------------	--

Description

Jointly estimates the Weibull scale parameter α and shape parameter β from an observed survivorship profile by minimizing MSE. It does not impose $\sum_x l_x m_x = 1$.

Usage

```
fit_weibull_free(
  lx_observed,
  fertility_rates = NULL,
  start_alpha = NULL,
  start_beta = NULL,
  alpha_bounds = c(1e-06, 1e+06),
  beta_bounds = c(1e-04, 100),
  control = list(),
  lx_tolerance = 1e-08
)
```

Arguments

lx_observed	Numeric observed survivorship profile.
fertility_rates	Optional fertility schedule of the same length as lx_observed.
start_alpha	Optional positive starting value for α .
start_beta	Optional positive starting value for β .
alpha_bounds	Positive lower and upper bounds for α .
beta_bounds	Positive lower and upper bounds for β .
control	Optional list passed to <code>stats::optim()</code> .
lx_tolerance	Positive tolerance used to validate lx_observed.

Details

This function is a descriptive reference comparison, not the default StablePopulation reconstruction route. In contrast with `select_beta()`, both parameters are free. If fertility_rates are supplied, they are used only after fitting to report the R_0 implied by the free profile.

The optimizer uses several starts on a log parameter scale and retains the best converged fit when available. Inspect attempts if convergence diagnostics or sensitivity to starting values are relevant.

Value

A list of class "stable_population_free_fit". Main elements are free alpha and beta, lx_fitted, MSE, RMSE, R0_fitted when fertility is supplied, a row-by-age profile, and attempts from all optimization starts.

See Also

`select_beta()` for the constrained observed-survivorship route.

Examples

```
lx_observed <- c(1, 0.92, 0.78, 0.57, 0.33, 0.12)
free_fit <- fit_weibull_free(lx_observed)
free_fit$alpha
free_fit$beta
```

normalize_fertility	<i>Normalize fertility using a reference survivorship profile</i>
---------------------	---

Description

Divides an age-specific fertility schedule by the net replacement calculated from a reference survivorship profile. The returned schedule satisfies $\sum_x l_x m_x = 1$ for that reference profile.

Usage

```
normalize_fertility(fertility_rates, lx_reference, tolerance = 1e-08)
```

Arguments

fertility_rates	Numeric vector of age-specific fertility rates.
lx_reference	Numeric reference survivorship profile. It must have the same length as fertility_rates.
tolerance	Positive numerical tolerance used to validate lx_reference.

Details

This helper makes fertility rescaling explicit before data enter a route that fixes $R_0 = 1$. It multiplies every fertility value by the same factor, preserving the age pattern while changing the schedule's overall scale.

It does not silently correct observed data or alter the package's internal demographic rule. Use it when the scientific analysis is deliberately framed at $R_0 = 1$ relative to a known reference survivorship profile. Raw and normalized schedules should be treated as distinct analytical inputs.

Value

A list of class "stable_population_normalized_fertility" containing the original fertility_rates, fertility_rates_normalized, R0_reference, the scaling factor, and the numerical check_R0.

See Also

[select_beta\(\)](#) for the observed-survivorship route and [run_reconstruction_excel\(\)](#) for workbook analysis. The Excel workflow uses supplied fertility values as-is, so normalized values must be prepared before saving them to a workbook.

Examples

```
mx <- c(0, 0, 0.30, 0.75, 0.60, 0.20)
lx <- c(1, 0.90, 0.75, 0.55, 0.30, 0.10)
normalized <- normalize_fertility(mx, lx)
normalized$check_R0
```

```
print.stable_population_demographic_profile
```

Print a demographic profile

Description

Print a demographic profile

Usage

```
## S3 method for class 'stable_population_demographic_profile'
print(x, ...)
```

Arguments

x	An object of class "stable_population_demographic_profile".
...	Additional arguments (currently ignored).

Value

x, invisibly.

```
print.stable_population_free_fit
```

Print a free Weibull fit

Description

Print a free Weibull fit

Usage

```
## S3 method for class 'stable_population_free_fit'
print(x, ...)
```

Arguments

x	An object of class "stable_population_free_fit".
...	Additional arguments (currently ignored).

Value

x, invisibly.

```
print.stable_population_normalized_fertility
```

Print a normalized fertility result

Description

Print a normalized fertility result

Usage

```
## S3 method for class 'stable_population_normalized_fertility'  
print(x, ...)
```

Arguments

x	An object of class "stable_population_normalized_fertility".
...	Additional arguments (currently ignored).

Value

x, invisibly.

```
print.stable_population_reconstruction
```

Print a stable population reconstruction

Description

Print a stable population reconstruction

Usage

```
## S3 method for class 'stable_population_reconstruction'  
print(x, ...)
```

Arguments

x	An object of class "stable_population_reconstruction".
...	Additional arguments (currently ignored).

Value

x, invisibly.

```
print.stable_population_scan
```

Print a stable population scan

Description

Print a stable population scan

Usage

```
## S3 method for class 'stable_population_scan'  
print(x, ...)
```

Arguments

x	An object of class "stable_population_scan".
...	Additional arguments (currently ignored).

Value

x, invisibly.

```
print.stable_population_selection
```

Print a stable population selection

Description

Print a stable population selection

Usage

```
## S3 method for class 'stable_population_selection'  
print(x, ...)
```

Arguments

x	An object of class "stable_population_selection".
...	Additional arguments (currently ignored).

Value

x, invisibly.

reconstruct_population

Reconstruct one constrained Weibull survivorship profile

Description

For a fixed Weibull shape parameter β and a fertility schedule m_x , calculates the corresponding α and reconstructs l_x under the StablePopulation condition $\sum_x l_x m_x = 1$.

Usage

```
reconstruct_population(
  fertility_rates,
  beta,
  tol = 1e-12,
  r0_tolerance = 1e-08
)
```

Arguments

fertility_rates	Numeric vector of age-specific fertility rates.
beta	Positive Weibull shape parameter.
tol	Positive numerical tolerance for root finding.
r0_tolerance	Positive tolerance used to assess numerical agreement with $R_0 = 1$.

Details

This is the fixed-beta route. Use it when beta has been selected elsewhere or when a specific beta value is being examined as a hypothesis. Age classes are generated internally as 0, 1, 2, ..., n - 1. The function uses the verified internal solver used by the extended workflow; it stops rather than silently treating a search endpoint as a valid root.

Value

A list of class "stable_population_reconstruction". Main elements are alpha, beta, lx, fertility_rates, lmx, numerical R0, residual, solver diagnostics, and a row-by-age table. The historical alias population is retained and equals lx; it is not the normalized stable age structure.

See Also

[scan_beta\(\)](#) to generate candidate profiles, [select_beta\(\)](#) to select a candidate using observed survivorship, and [derive_demographic_profile\(\)](#) to calculate R, D, and B.

Examples

```
mx <- c(0, 0, 0.30, 0.75, 0.60, 0.20)
reconstruction <- reconstruct_population(mx, beta = 1.10)
reconstruction$alpha
reconstruction$table
```

run_analysis

Run the Legacy Source-Tree Excel Workflow

Description

This legacy function reads fertility rates and one pre-defined beta value from each worksheet of `inst/extdata/Input_Data.xlsx`. It calculates the corresponding alpha under $R_0 = 1$ and writes one results workbook per worksheet/species.

Usage

```
run_analysis()
```

Details

The historical Excel layout is fixed:

- column B contains the fertility schedule m_x , excluding its header;
- cell C2 contains the pre-defined Weibull shape parameter beta;
- the first row is interpreted as a header row.

This function is intentionally retained as a legacy source-tree workflow for compatibility with StablePopulation 1.0.3 and with the workflow described in the ecoinformatics note. It must be run from a Git checkout or unpacked source archive that contains `inst/extdata/Input_Data.xlsx`. The historical workbook is excluded from built package artifacts, and installed package libraries are generally not writable, so this is not a supported installed- package workflow.

For new analyses, use `run_reconstruction_excel()`, which accepts an external workbook and writes results to a user-selected or automatically derived location.

Value

No return value. Called for side effects: reading the historical source-tree workbook, writing one result workbook per worksheet in `inst/extdata`, and printing messages.

See Also

[excel_sheets](#), [read_excel](#), [createWorkbook](#), [addWorksheet](#), [writeData](#), [saveWorkbook](#)

run_reconstruction_excel

Reconstruct stable-population profiles from an Excel workbook

Description

Reads one or more Excel sheets containing a fertility schedule and an optional observed-survivorship profile. The default workbook is designed for direct use: it starts with an overview and then provides one result sheet per processed input sheet. A full audit workbook can be requested when candidate and profile diagnostics are needed.

Usage

```
run_reconstruction_excel(
  input_file = NULL,
  output_file = NULL,
  mode = c("auto", "scan", "select", "fixed"),
  beta_values = seq(0.05, 3, by = 0.05),
  terminal_window = NULL,
  sheets = NULL,
  skip_non_data = TRUE,
  age_column = NULL,
  fertility_column = NULL,
  survivorship_column = NULL,
  beta_column = NULL,
  tol = 1e-12,
  r0_tolerance = 1e-08,
  output_detail = c("standard", "full")
)
```

Arguments

input_file	Optional path to an existing Excel workbook. When omitted or NULL in an interactive R session, a file chooser opens. It must be supplied in non-interactive use.
output_file	Optional path for the results workbook. When omitted, creates <input name>_StablePopulation.xls beside the input file.
mode	One of "auto", "scan", "select", or "fixed". With "auto", the function uses "select" when observed survivorship is present, "fixed" when a recognized beta column contains one positive value, and "scan" otherwise.
beta_values	Positive Weibull shape values to scan.
terminal_window	Optional terminal-survivorship window for the scan route.
sheets	Optional workbook sheet names to process. By default, all sheets are inspected.
skip_non_data	Logical. When TRUE and sheets is omitted, sheets without a recognized fertility column are recorded as skipped instead of stopping the run.
age_column	Optional explicit age-column name.
fertility_column	Optional explicit fertility-column name.
survivorship_column	Optional explicit observed-survivorship column name.
beta_column	Optional explicit fixed-beta column name.
tol	Positive numerical tolerance for root finding.
r0_tolerance	Positive numerical tolerance for the $R_0 = 1$ check.
output_detail	One of "standard" or "full". The default "standard" output contains only Overview and one Result_<sheet> worksheet per processed input sheet. The "full" output additionally contains Candidates_<sheet> for select and scan routes, Profiles_<sheet> for scan routes, and a final Metadata worksheet.

Details

Headings are matched case-insensitively and may include descriptive text or units. For example, Age (years), mx (Fertility Rate), and lx (Survivorship) are recognized automatically. Fertility aliases include mx, m_x, fertility, fertility_rates, fecundity, and fecundidad. Observed-survivorship aliases include lx, l_x, lx_observed, l_x_observed, survivorship, survival, and supervivencia. Age aliases include age, edad, age_class, and clase_edad. Fixed-beta aliases include beta, weibull_beta, shape, and shape_parameter.

A recognized age column is retained as an output label, whereas all calculations use the internal consecutive class index 0, 1, ..., n - 1. Fully blank rows are ignored. Missing or non-numeric fertility values inside a data table stop the run with the affected Excel row numbers; they are never silently removed.

A select or fixed-beta route produces one reconstructed profile. A scan route does not choose one arbitrary beta: its result sheet lists the stable candidates, or terminal-window-admissible candidates when a terminal window is supplied.

The Excel workflow analyses fertility exactly as supplied; it never applies implicit fertility normalisation. When a schedule must be rescaled relative to a reference survivorship profile, use [normalize_fertility\(\)](#) in R first and save the resulting fertility values to the workbook.

Value

Invisibly, a list with output_file, one result object per processed input sheet, and a metadata table that also records skipped sheets. Metadata is returned regardless of output_detail.

Examples

```
## Not run:
# Opens a file chooser in an interactive R session and creates the
# standard output workbook.
if (interactive()) {
  run_reconstruction_excel()
}

# Creates demography_StablePopulation.xlsx beside demography.xlsx.
run_reconstruction_excel("demography.xlsx")

# Request technical candidate and profile diagnostics.
run_reconstruction_excel("demography.xlsx", output_detail = "full")

# Use explicit column names when an input workbook has custom headings.
run_reconstruction_excel(
  input_file = "demography.xlsx",
  output_file = "results.xlsx",
  sheets = "Ovis_dalli",
  age_column = "Age class",
  fertility_column = "Female fertility",
  survivorship_column = "Observed survivorship",
  beta_column = "Beta"
)

## End(Not run)
```

scan_beta

*Scan Weibull beta values under the stable-population constraint***Description**

For every candidate β , calculates the corresponding α , reconstructs a Weibull survivorship profile l_x , and checks whether the numerical result satisfies $R_0 = \sum_x l_x m_x = 1$.

Usage

```
scan_beta(
  fertility_rates,
  beta_values = seq(0.05, 3, by = 0.05),
  tol = 1e-12,
  r0_tolerance = 1e-08,
  terminal_window = NULL
)
```

Arguments

fertility_rates	Numeric vector of non-negative age-specific fertility rates.
beta_values	Positive Weibull shape values to scan.
tol	Positive numerical tolerance for root finding.
r0_tolerance	Positive tolerance for the numerical R_0 check.
terminal_window	Optional numeric vector c(lower, upper) in $[\emptyset, 1]$. A stable profile is terminally admissible when its final l_x lies in this inclusive interval. Default: NULL.

Details

Age classes are always assigned internally as $\emptyset, 1, 2, \dots, n - 1$. This is the scenario route when observed l_x is unavailable: it generates a family of constrained profiles and does not select an empirical optimum.

When terminal_window is supplied, the function marks profiles whose final survivorship lies inside that inclusive interval. The terminal criterion filters candidate scenarios after the numerical R_0 check; it does not estimate beta. When at least one candidate passes the window, terminal_extremes\$first and terminal_extremes\$last are the first and last admissible candidates in ascending beta order, preserving the historical Model-V2022 scenario convention. They should not automatically be interpreted as minima or maxima of every downstream ecological quantity.

Value

A list of class "stable_population_scan". Main elements are summary (one row per beta, including numerical diagnostics and terminal admissibility), profiles (one survivorship column per beta), admissible_summary, profiles_admissible, and terminal_extremes when an active terminal window retains at least one profile. checks records the numbers of scanned, stable, and admissible candidates.

See Also

[select_beta\(\)](#) for the route with observed survivorship, [reconstruct_population\(\)](#) for one fixed beta, and [derive_demographic_profile\(\)](#) for derived demographic quantities.

Examples

```
mx <- c(0, 0, 0.30, 0.75, 0.60, 0.20)

# All stable candidate profiles
scan <- scan_beta(mx, beta_values = seq(0.05, 1.50, by = 0.05))

# Scenario route without observed lx
terminal_scan <- scan_beta(
  mx,
  beta_values = seq(0.05, 1.50, by = 0.05),
  terminal_window = c(1e-4, 0.05)
)
terminal_scan$admissible_summary
```

select_beta

*Select a constrained Weibull beta using observed survivorship***Description**

Generates Weibull survivorship candidates under $R_0 = 1$ and selects the candidate with the smallest RMSE against an observed survivorship profile l_x . It is not a free two-parameter Weibull fit: α remains determined by the fertility schedule and the stability constraint for every candidate β .

Usage

```
select_beta(
  fertility_rates,
  lx_observed,
  beta_values = seq(0.05, 3, by = 0.05),
  tol = 1e-12,
  r0_tolerance = 1e-08,
  lx_tolerance = 1e-08
)
```

Arguments

fertility_rates	Numeric vector of non-negative age-specific fertility rates.
lx_observed	Numeric observed survivorship profile. It must have the same length as fertility_rates, start at one, and be non-increasing.
beta_values	Positive Weibull shape values to scan.
tol	Positive numerical tolerance for root finding.
r0_tolerance	Positive tolerance for the numerical R_0 check.
lx_tolerance	Positive tolerance used to validate lx_observed.

Details

This is the route to use when observed survivorship is available. The function scans the supplied beta grid and returns the best grid candidate; inspect `results` rather than treating the selected value as a continuous unconstrained optimum. If the selected candidate lies at the lower or upper edge of the scanned beta grid, the function issues a warning and records a boundary note in the returned object, because the best value may lie outside the supplied range.

Fertility is analysed exactly as supplied. The function never rescales it implicitly. When the methodological decision is to express an observed fertility schedule relative to the stable condition, call `normalize_fertility()` first and supply `fertility_rates_normalized`. Using raw fertility and using an explicitly normalised schedule are different analyses and should remain distinguishable.

Value

A list of class "stable_population_selection". Main elements are `best_beta`, `best_alpha`, `best_lx`, `best_R0`, `best_MSE`, `best_RMSE`, `best_profile` (age-by-age comparison), `results` (all scanned candidates), `beta_at_boundary`, `beta_boundary_side`, `beta_boundary_note`, and the underlying scan object.

See Also

`scan_beta()` for the route without observed survivorship, `normalize_fertility()` for explicit fertility rescaling, and `fit_weibull_free()` for an unconstrained reference fit.

Examples

```
mx <- c(0, 0, 0.30, 0.75, 0.60, 0.20)
lx_observed <- c(1, 0.8302461, 0.6893086, 0.5722958, 0.4751464, 0.3944885)

selection <- select_beta(mx, lx_observed)
selection$best_beta
selection$best_profile
```

StablePopulation

StablePopulation: constrained Weibull survivorship profiles

Description

StablePopulation reconstructs discrete Weibull survivorship profiles under a stable and stationary population assumption. Its core constraint is $R_0 = \sum_x l_x m_x = 1$, where l_x is survivorship and m_x is age-specific fertility. All calculations use consecutive internal age-class indices 0, 1, 2, ..., n - 1.

Details

The package is the species-level demographic layer of a wider modelling workflow. It reconstructs survivorship and derives a stable age structure and exit-by-death profile. It does not yet calculate body-mass-weighted biomass, aggregate prey communities, or distribute resources among predators.

Choosing a reconstruction route

- Use `reconstruct_population()` when a single Weibull β is already specified.
- Use `select_beta()` when observed l_x is available. It scans candidate β values, solves α under $R_0 = 1$ for each candidate, and selects the smallest RMSE.
- Use `scan_beta()` when observed l_x is unavailable. It returns candidate scenarios; an optional terminal-survivorship window can retain profiles compatible with an explicit final-class criterion.
- Use `fit_weibull_free()` only as a descriptive free two-parameter comparison. It does not impose $R_0 = 1$.

Fertility normalisation

`normalize_fertility()` explicitly rescales a fertility schedule relative to a reference survivorship profile. It preserves the fertility pattern while changing its overall scale so that the reference R_0 equals one. Normalisation is never performed silently by `select_beta()` or `run_reconstruction_excel()`. Use it only when that rescaling is the stated analytical choice.

Derived demographic quantities

`derive_demographic_profile()` calculates the normalized stable structure $R_x = l_x / \sum_x l_x$, the raw exit-by-death profile $D_x = R_x - R_{x+1}$ with $D_n = R_n$, its relative version $D_x / \sum_x D_x$, and conditional survival $B_x = l_{x+1} / l_x$. The raw D profile and the relative D profile are distinct objects.

Excel workflows

`run_reconstruction_excel()` is the recommended external-workbook workflow. In mode = "auto", a sheet with observed survivorship uses the selection route; without observed survivorship, a one-value beta column uses the fixed-beta route, and otherwise the scan route is used. The historical no-argument `run_analysis()` function is retained only for compatibility with the StablePopulation 1.0.3 source-tree workbook layout.

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References

Martín-González, J. A., Rodríguez-Gómez, G., and Palmqvist, P. (2019). Survival profiles from linear models versus Weibull models: Estimating stable and stationary population structures for Pleistocene large mammals. *Journal of Archaeological Science: Reports*, 25, 119–127.

 validate_fertility_rates

Validate age-specific fertility rates

Description

Checks that an age-specific fertility schedule is a finite, non-negative, numeric vector with at least one positive value. StablePopulation assigns age classes internally as consecutive indices 0, 1, 2,

Usage

```
validate_fertility_rates(fertility_rates, min_length = 2L)
```

Arguments

fertility_rates	Numeric vector of non-negative age-specific fertility rates.
min_length	Minimum permitted number of age classes.

Value

A validated numeric vector.

 validate_lx_observed *Validate an observed or reconstructed survivorship profile*

Description

Checks that a survivorship vector l_x is finite, one-dimensional, bounded between zero and one, starts at one, and is non-increasing with age. Tiny rounding deviations within tolerance are accepted and then clipped to the interval $[0, 1]$.

Usage

```
validate_lx_observed(
  lx_observed,
  expected_length = NULL,
  tolerance = 1e-08,
  min_length = 2L
)
```

Arguments

lx_observed	Numeric vector of survivorship values.
expected_length	Optional expected length.
tolerance	Positive numeric tolerance for validation.
min_length	Minimum permitted number of age classes.

Value

A validated numeric survivorship vector.

weibull_survival	<i>Weibull function for the survival rate</i>
------------------	---

Description

This function calculates the survival rate to reach a specific age using the Weibull function.

Usage

```
weibull_survival(alpha, beta, age)
```

Arguments

alpha	A numeric value representing the scale parameter of the Weibull distribution.
beta	A numeric value representing the shape parameter of the Weibull distribution.
age	A numeric value representing the age.

Value

A numeric value giving the survival rate (probability) for reaching the given age.

Examples

```
weibull_survival(1.5, 0.8, 10)
```

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